SCHOOL OF ECONOMICS AND MANAGEMENT

Women in the Swedish Video Game Industry

2006-2020

Underrepresentation, Education, and Occupational Segregation

by

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Abstract

This thesis investigates the underrepresentation of women in the Swedish game industry from 2006 to 2020. Globally, women in video game development constitute a small fraction of the workforce, often occupying non-developmental roles. Using a mixed-methods approach, this study explores the relationship between education, occupational segregation, and the overall underrepresentation of women in the game industry. Regression analyses reveal women's increasing interest in video game development and demonstrate how educational pathways and approaches influence workforce diversity. The study also examines the presence of occupational segregation in the Swedish game industry and its broader implications. Recommendations for future research on the topic are discussed.

Keywords: underrepresentation, gender diversity, video game development, occupational segregation, education

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1. Introduction

The video game industry is regarded as one of the most dynamic and fast-growing industries in th global economy, characterized by its profound economic, social, and cultural impact (Ecorys & KEA, 2023). Since its first steps in the 1960s, video games have been primarily marketed towards boys and young men, reinforcing gender-specific stereotypes and perpetuating certain cultural norms (Lhuillery, 2020). Similarly, the workforce within the video game industry has traditionally been male dominated, with women consistently underrepresented in various roles, particularly in leadership and technical positions (Ecorys & KEA, 2023). The underrepresentation of women in the video game industry is a complex issue that encompasses numerous ongoing barriers. These barriers include unequal opportunities for career advancement, biased perceptions of women's capabilities, and systemic challenges within organizational and societal structures (Bailey et al., 2021).

The conversation around gender diversity in the video game industry has steadily gained momentum. As noted by advocacy organizations such as Women in Games (2023), addressing the underrepresentation of women in this sector is not merely about increasing their numerical presence, but it also involves critically examining and improving the nature of roles and opportunities available to women.

This thesis aims to investigate the state of gender diversity within the Swedish video game industry, focusing on three interconnected areas: the historical and current share of women in the workforce, the relationship between education and women's representation, and the extent and implications of occupational gender segregation. To achieve this, the research is structured around the following key questions:

Research Question 1.

How has the share of women changed over time in the Swedish game industry?

Research Question 2.

What is the relationship between education and women's underrepresentation in the video game industry?

Research Question 3.

To what extent is occupational segregation present in the video game industry and what are its implications?

Firstly, Sweden is recognized as an equal country with high gender diversity numbers (Stanfors, 2014) and a significant presence in the gaming sector. In 2020, Sweden accounted for 24.5% of the European Union's total video game companies, highlighting its importance in the international market (Ecorys & KEA, 2023). Given its increasing importance and role in the global markets, gender diversity in the Swedish video game industry has been addressed by the trade organization Dataspelsbranschen. Dataspelsbranschen (2012) has noted a strong focus and active work on gender equality in the industry, and their 2019 Game Developer Index Report highlights progress, celebrating an increase in female employees in the industry. However, the extent of women's underrepresentation in the Swedish video game industry over time is yet to be depicted.

Education is a critical factor in shaping the skills and qualifications of the workforce, and it plays a vital role in addressing gender disparities. As highlighted by Dataspelsbranschen (2023), higher education programs aimed at the gaming industry are to developing the skills necessary for the industry's workforce. However, women remain underrepresented in STEM fields, including game development, which falls under the broader STEM umbrella (Lhuillery, 2020). This thesis seeks to explore the extent of this underrepresentation and its implications for the industry. By employing multiple regression analysis and linear regression scatter plots, the second section of the *Results & Analysis* will examine the relationship between female applications, students accepted, and employees in the industry.

As the video game industry's size and impact have grown, it has opened new employment opportunities for a variety of occupations over the years within art, technology, and business, all of them demanding specific skills and experience (Dataspelsbranschen 2009; Ecorys & KEA, 2023). The underrepresentation of women in technical positions is a social issue commonly known as *occupational segregation*, one of the challenges women may encounter in their professional

careers (Harvey and Fisher, 2015). Miller et al (2004) argue that occupational segregation is one of the strongest influences in young people's careers, and as such, I find it important to understand the changes in women's participation in the video game industry and whether the Swedish game industry experiences occupational segregation. The third section of the *Results & Analysis* will discuss the extent of occupational segregation in Sweden, and within the Swedish game industry and its broader implications.

This research aligns with broader efforts to create a more equitable and diverse professional landscape within the video game industry and beyond. Firstly, recognizing and understanding the changes in diversity in various sectors within the industry is crucial in ensuring the full participation of women in society (Directorate-General for Research and Innovation, 2022). The results of this thesis are relevant to the ongoing discussion on gender diversity by the Nordic Council, the European Commission, and organizations at the national level, which frequently acknowledge progress or growth. According to the European Commission (2023), inclusion and diversity have improved in the video game sector, but there are still improvements to be made. This thesis aims to provide suggestions based on the results of the research to further address women's underrepresentation in the video game industry.

Finally, it is important to note that this research focuses exclusively on individuals who identify as male or female, excluding non-binary perspectives due to scope limitations. Future studies could expand this scope to encompass a broader spectrum of gender identities, providing a more holistic understanding of diversity in the gaming industry.

2. Theoretical Approach and Previous Research

Since its first steps in the 1960s, video games have been primarily marketed towards boys and young men and has been seen as a "masculine" hobby (Lhuillery, 2020). This gendered marketing approach has historically contributed to the underrepresentation of women in the creation and development of video games. As Lhuillery (2020) highlights, female developers accounted for only 3% of the workforce in 1989. Although women represent nearly half of the international video game players, women continue to experience gender discrimination and sexism both as players and developers (Bailey et al, 2019).

Harvey and Fisher (2013) are two scholars who have shared the results of a 2001-2011 analysis, revealed that men constituted, on average, 90% of the video game industry's workforce. More recent research by Bailey et al. (2021) highlights that the ratio of women to men in development teams varies from 1:6 to 1:17 underscoring the persistent gender disparity. Research by Wey et al. (2016) and Vedres & Vásárhelyi (2023) emphasizes the value of gender diversity in fostering innovation and creativity within the industry. A diverse workforce enhances the industry's capacity to design games that resonate with a wider audience by reflecting the diversity of its consumers.

Weststar and Marie-Josée (2018) outline the career trajectory of women in game development through three critical stages: exposure, education, and job experience. Education, in particular, plays a pivotal role in addressing skill gaps and expanding participation across various roles in the industry, including development, production, design, art, and testing (Bogg & Prescott, 2011). Apart from formal education provided by national universities, Harvey and Fisher (2015) explore alternative pathways, such as self-taught programming and other skills that offer similar career paths to individuals interested in joining the video game industry. Do-it-yourself (DIY) game design and self-taught programming in particular offer broader opportunities by addressing gaps in computer literacy (Yasmin B. Kafai and Kylie Peppler 2011, cited in Harvey and Fisher, 2015).

However, women pursuing education and careers in STEM fields often face systemic barriers. Research by Clark et al. (2021) highlights the prevalence of sexist academic environments, which can deter women from completing STEM degrees. Similarly, Blickenstaff (2005) and Haines (2004) observe that such environments may lead women to switch to non-technical disciplines before graduation. This imbalance in higher education translates into a disproportionately male workforce, perpetuating a cycle of underrepresentation (Miller et al., 2004). Similarly, a homogeneous workforce can impact women's career choices as pointed by Clark et al (2021), as women may feel less confident about following a STEM career knowing that there is an unequal number of job opportunities compared to their male counterparts.

Lemons and Parzinger (2001) have researched the over-concentration of women in jobs such as household workers, service employees, elementary school teachers, secretarial workers, and nurses. In comparison, as share by Blickenstaff (2005) women are underrepresented in science and technical jobs. This phenomenon is known as *occupational gender segregation*, the

underrepresentation of women in certain occupations that persists despite the growth of female participation in the labor market overall (European Institute for Gender Equality, 2024; Bogg & Prescott, 2011). For instance, Mandel and Semyonov (2006) share that when women are segregated into certain jobs, those jobs usually pay less, have fewer opportunities for progression, less job autonomy, and less authority within the organization. Moreover, gender segregation restricts interactions between men and women in the workplace, potentially reinforcing stereotypes (Bogg & Prescott, 2011).

The video game industry comprises different roles including development, production, design, art, and testing (Bogg & Preston, 2011). In their research paper, Bailey et al (2019) cover the differences in gender representation by roles, reaching the same conclusion as Bogg and Preston (2011) that women remain under-represented in technical and developing positions such as programming and design, whilst over-represented in non-developmental or managerial positions such as marketing. By interviewing women in the United Kingdom who worked in the industry, Haines (2004) found that 73% of the female participants worked in non-developmental positions. My Nilsson (2021) researched the occupational distribution across different occupations in the Swedish game industry in 2021. Nilsson highlights the limited data on occupational distribution within the Swedish video game industry. To overcome these challenges, My Nilsson's (2021) research is based on observations through LinkedIn, in which the researcher has pointed to the reliability and limitations of their findings. Her research points to a possible research gap in occupational gender segregation in the industry which this paper investigates uncovering.

According to Lemons and Parzinger (2001), the root of occupational gender segregation lies in traditional models and internalized social values in which women are taught from a young age to be nurturers and caregivers. This early socialization shapes perceptions and career aspirations, particularly in fields such as technology and science, where women tend to hold more negative preconceptions than men, leading to a lack of interest in these sectors (James & Cardador, 2007, cited in Boggs & Prescott, 2011). Stereotypical beliefs about women's competence in STEM careers persist in the workplace, influencing career choices and perpetuating segregation (Clark et al., 2021). The consequences of occupational segregation spread into skill provision, as occupational segregation is one of the strongest influences on people's career choices (Miller et

al, 2004). Miller et al (2004) share that individuals perceive an occupation or career as “male” or “female” based on their gender distribution associating that career with specific gender-based skills.

Women encounter barriers throughout their career progression in the video game industry, from recruitment to advancement. Weststar and Marie-Josée (2018) describe these challenges across exposure, education, and job experience. Blickenstaff (2005) proposes solutions to address gender disparities, such as equal access to educational resources and the elimination of sexist language and imagery. While these suggestions are valuable, further research is needed to evaluate their effectiveness.

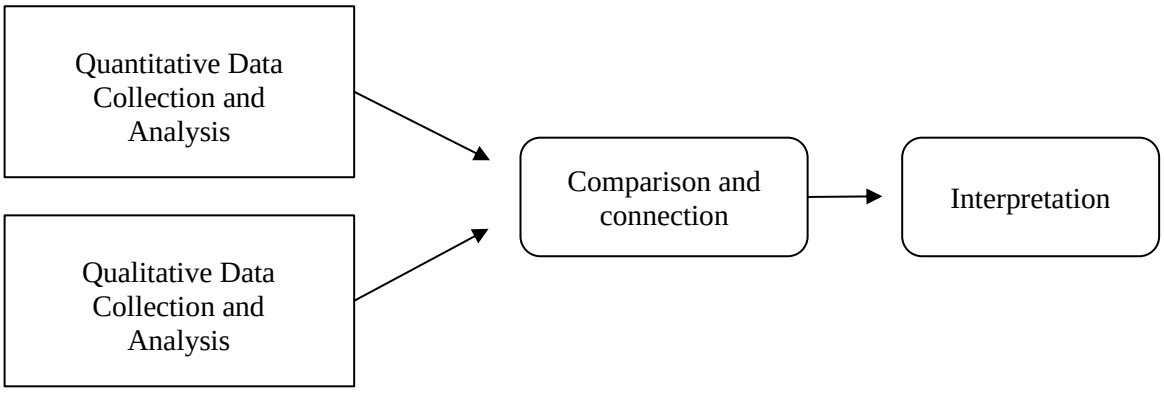
The challenges faced by women during the recruitment processes are an important point of discussion that should be investigated in a deeper analysis. For example, the Statistical Discrimination Model would be significant as it argues that due to the limited information recruiters have on the candidate's productivity, recruiters are then likely to discriminate against “lower average expected productivity” to reduce training costs and increase overall productivity (Mandel & Semyonov, 2006). Another relevant theory that has not been utilized in this research is the presence of a glass ceiling within the video game industry. Evidence has shown that in typically male occupations there are increased opportunities for advancement and career progression than in predominantly female occupations (Tomaskovic-Devey, 1993, in Hultin 2003). Further, women’s decision to be part of the video game industry may be affected by working conditions including the gender wage gap and a sexist work culture reported in international game studios (Bailey et al, 2019). It would be interesting to study the extent of a sexist culture in Swedish game studios and whether it has had an impact on women’s representation. Similarly, research on Sweden’s video game industry gender wage gap is yet to be carried on. This thesis acknowledges the need for further research to complement this study.

3. Data and Methods

This section presents the thesis methodology, beginning with the research design in the form of a convergent mixed-methods approach depicted in Diagram 1 below. The mixed methods complement each other acting as mutual explanatory tools for the stated hypotheses and

minimizing limitations. For instance, the mixed-method approach will help understand and interpret the quantitative findings in a better way given the small quantitative sample. Similarly, the quantitative results help interpret broader findings in the qualitative sources. The integration process occurred during the analysis and presentation of the results in the upcoming sections.

Diagram 1. Convergent Parallel Design of this Research



This research will focus on Sweden due to a variety of factors. Firstly, given that Sweden is one of the most equal countries in Europe would reinforce the main hypothesis of women's underrepresentation in the video game industry even in countries with a strong support system like Sweden. Secondly, Sweden was selected due to the availability and access to public information and statistics, and the use of English as one of their main languages facilitating data collection. Thirdly, Sweden is home to game development companies of different scales; from independent studios to world-renowned developers, such as *King*. The time 2006 - 2020 was selected given that the earliest data available is from 2006, whilst the end period of 2020 was selected to avoid the impact of the COVID-19 pandemic.

3.1. Qualitative Methods and Sampling

For the qualitative analysis of this paper, I will be analyzing secondary data obtained from various published reports by non-governmental organizations, governmental organizations, private companies, and regional bodies. These sources were selected for their relevance to this research topic.

The report *Building a Fair Playing Field* (2023) by Women in Games was used to get an overview on the status of women in the international video game industry as both players and developers. Reports under the European Commission's research body include *She Figures* (2018) and *Understanding the Value of a European Video Games Society Final Report* (2013), providing a synopsis on Sweden as member of the European Union. The *Nordic Game Education Index* (2017) report by the Nordic Council and the *Swedish Game Education: 2001-2016* (2017) report from the project *GameHub Scandivania* were used to provide information on Sweden as a Nordic country. Various reports from Dataspelsbranschen were used, including the yearly *Game Developer Index* reports from 2006 to 2020, *Talent, Education and the Art of Making Games* (2023) and *Paving Ways to the Games Industry* (2024). Finally, the company report *Environmental, Social, and Governance Report* (2020) from Activision Blizzard was used for information on the mobile developer company *King* and the *Developer Satisfaction Survey Summary Report* (2016) by the International Game Developer Association (IGDA) provided data on the video game industry workforce as a whole.

To ensure a comprehensive understanding of the subject, the reports were critically examined and cross-verified with each other. Findings on the topic of gender diversity and gender distribution were identified to establish patterns and differences to be later connected with the quantitative results of this research.

These reports were scrutinized, as they are second-hand sources that come with limitations. Dataspelsbranschen has shared their awareness and concern on the level of transparency of developer studios and the submitted reports that provide the information for the *Game Developer Index* reports. This limits the reliability of the reports, as it compiles information that the studios have chosen to share, which could be biased. Moreover, as stated by Dataspelsbranschen themselves, smaller studios and solo developers are challenging themselves to keep track of, as they might not be registered as a company and might not be obliged to provide annual reports. Similarly, the reports from non-profit organizations such as Women in Games (*Building a Fair Playing Field*) offer both first-hand data from interviews and second-hand data from governmental and non-governmental organizations (including Dataspelsbranschen). These types of reports may emphasize certain perspectives that align with their advocacy goals, thus limiting their reliability.

Reports from governmental organizations such as the European Commission and Nordic Council were valuable for the analysis, however, the researcher acknowledges their potential bias as they might reflect the priorities and policies of such organizations.

To increase the reliability and validity of the sources, I used reports from multiple authors to get different perspectives on gender diversity and women in the industry. Including reports from different years allowed for a clearer timeline providing insight into potential gaps. Finally, the findings were interpreted within the context of each report's origin, purpose, and intended audience.

3.2. Quantitative Methods and Sampling

The data used to run the statistical analysis was retrieved from existing datasets and reports via purposive (or judgmental) sampling. In this case, the data selected was based on specific criteria to obtain the most relevant information for this research. For instance, Statistikmyndigheten SCB (Sweden's Statistical Database) served as the primary source regarding employees per occupation between the years 2014 and 2020. The data provided by the search was compiled on a spreadsheet from which five occupations were selected. These occupations were selected via purposive sampling to have occupations with a high proportion of women, and STEM-related occupations.

Similarly, for the education section, I created a small-sized database split into two datasets using first-hand data provided by *Universitets- och högskolerådets* (UHR) dataset using a purposive approach. In this case, the data was narrowed to those Autumn programs between 2008 and 2019 that were aimed at the video game industry by searching for the word "*spel*", the Swedish word for "game". The results were then exported to an Excel file where the program titles were translated from Swedish to English using direct Google Translate and relying on the program syllabus when the direct translation was unclear. The search resulted in a small database (summary of two data sets shown in Appendix A), composed of a total of 759 observations, and eight variables, split into categorical and numerical:

Categorical variables

Program Title *program_id*

University	*school*
Numerical variables	
Female Applications	*female_applications*
Male Applications	*male_applications*
Total Applications	*total_applications*
Year	*year*
Women accepted	*accepted_female*
Men accepted	*male_accepted*

Appendix B shows the code used for quantitative analysis with the software STATA, where Chat GPT was used as a support tool to correct syntax and grammatical errors within erroneous STATA code. A general univariate analysis of the continuous variable *female_applications* is done as an explorative study to produce statistics and measures of central tendency (mean, minimum, maximum, and sum). To further deepen the discussion, a multiple linear regression analysis was run to find the relationship between female employees and women accepted to high-education programs (*accepted_female* and *year* as independent variables). A multiple regression analysis was chosen over a simple linear regression to account for the influence of time. The researcher further acknowledges the importance of time-based factors that this model does not consider, such as the length of the educational programs, which require a more complex and thorough analysis.

The researcher acknowledges the limitations of the sampling method used for this study. As the purposive sampling used is selective, it does not produce a representative sample of the entire population, which may limit generalizability and bias. Using the word "*spel*" may limit the number of programs being looked at, for a more rounded study one might choose to do a manual selection based on the program syllabus. It is also noted that between 2011-2012, some programs with the prefix "*spel"* did not provide the competencies needed for game development as shared by Björn Berg (2017). Further, there are many Higher Vocational Education (HVE) focused on game development; however, these have not been considered for this study, limiting the applicability of

the results. It is also worth noting that the *Nordic Game Education Index* (2017) has data on programs related to game development. These numbers, although similar to the results obtained by the researcher's search and method, were excluded to ensure continuity in the analysis done.

3.3.Limitations

The most significant limitation of this study lies in the lack of comprehensive data and publicly available information regarding occupational distribution within the Swedish video game industry. Game developer studios have not consistently included detailed occupational data in their annual reports, press releases, or web pages accessible to the public. This omission creates challenges in the full understanding of the occupational distribution by role in the workforce and its implications. It is unclear if this stems from privacy concerns or a lack of prioritization to measure and report this metric. Regardless of the reason, the scarcity of occupational information represents a major obstacle to gaining a comprehensive understanding of the workforce composition. Although this research was conducted with the limited data available on occupational distribution, it is important to emphasize that the analysis do not fully capture the Swedish video game industry. This gap in data limits the depth and accuracy of the findings.

To maintain the scope and simplicity of this study, several factors have been excluded from the analysis. For instance, demographic variables such as age, marital or civil status, number of children, citizenship, and the nature of the employment arrangement (e.g., hybrid or remote work) were not considered. Understanding how and to which extent these personal and professional factors may impact a woman’s career is key to getting a deeper insight to further improve gender diversity. Similarly, this study does not explore non-developmental educational programs, such as those focused on business management, marketing, etc., and employment in the video game industry. Although not associated with game development, these programs may still lead to the game industry. Their exclusion limits the scope of the analysis and reduces its applicability for understanding the range of career opportunities available within the industry. Thus, future research on gender distribution that includes these educational paths should be conducted.

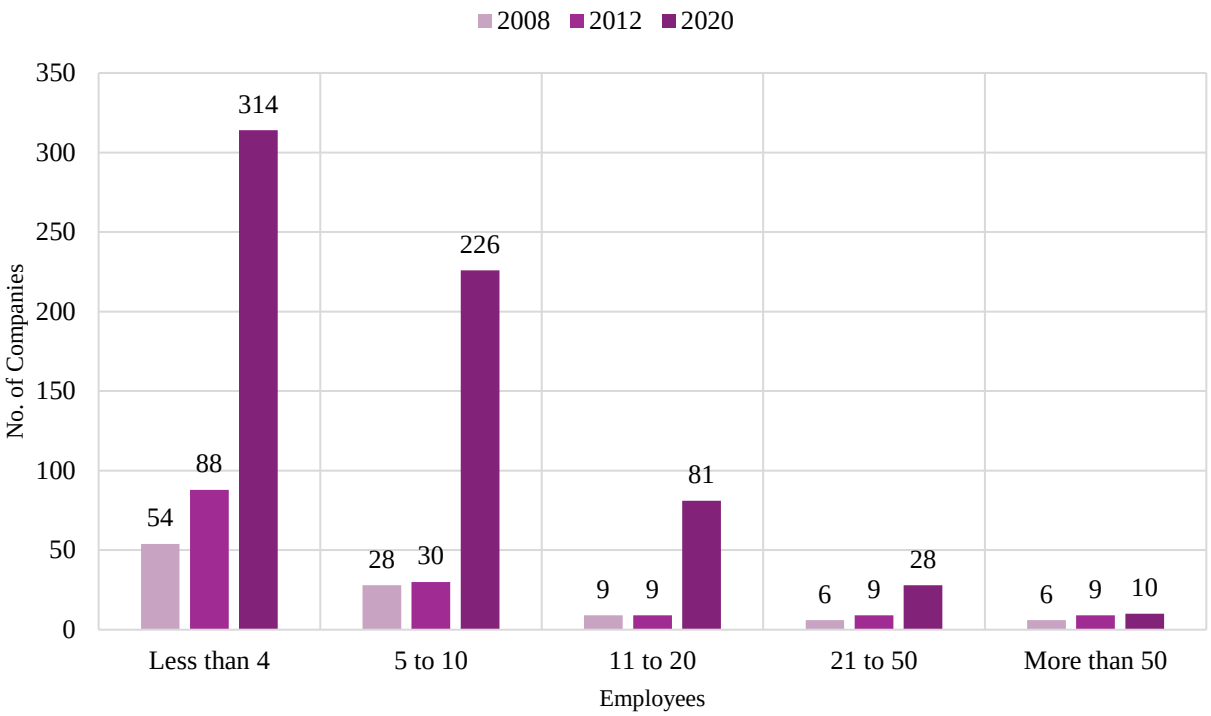
4. Results & Analysis

This section presents the analytical framework and the results of this research. It is divided in three parts following the three main research questions. The first part (4.1) dives into the general trends of the Swedish game industry and women's representation. The second part (4.2) covers the role of higher education in the provision of skills to promote women entering the video game industry. This section includes the results of the qualitative analysis described in the methods section. The last part of this section (4.3) explores occupational distribution in Sweden and the extent of occupational segregation in the video game industry.

4.1. Women in the Swedish Video Game Industry

The Swedish video game industry has experienced substantial growth over the past few decades, marked by a significant rise in the number of companies. In 2006, Sweden had just 68 registered video game companies, which skyrocketed to 667 by 2020. This dramatic increase underscores the growing population of video games and its industry within Swedish society. An increase in independent studios has characterized this rise in the number of companies. Figure 1 highlights the sharp increase in independent or small-sized studios with less than four employees, from 54 in 2008 to 314 by 2020. Similarly, studios composed of five to ten employees also increased significantly from 28 in 2008 to 226 by 2020. This trend could suggest an increase in people's interest in becoming game developers, where they choose to become solo developers or join smaller independent studios. Figure 1 also shows the growth and expansion of medium-sized companies in Sweden, with the number of companies employing between eleven and fifty people increasing over time. In contrast, there was minimal growth among larger companies, with those employing over 50 people rising from six in 2006 to ten in 2020. However, despite this minimal increase in company numbers, it is important to remark that larger companies still saw significant growth. For example, *EA Digital Illusions* (*DICE*) was composed of 240 employees in 2008 to an impressive 794 employees by 2019 (Dataspelsbranschen, 2020). Similarly, mobile developer *King* increased their total number of employees from 152 in 2012 to 718 in 2019.

Figure 1. Swedish Video Game Companies by Employee Size

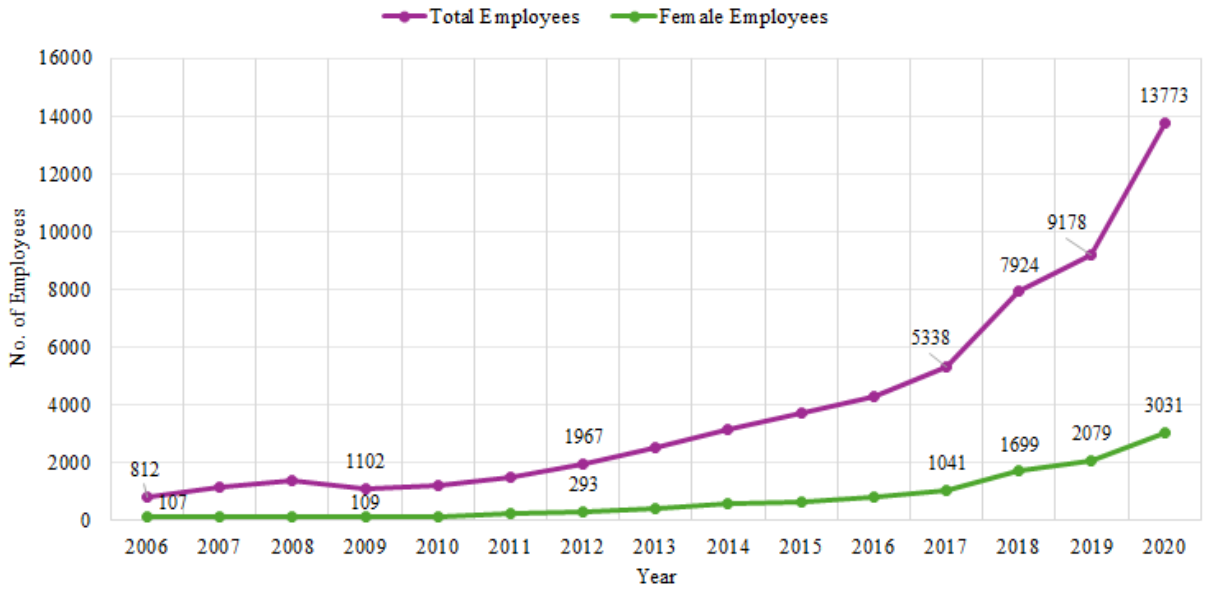


Source: Dataspelsbranschen

The growth in the Swedish video game industry came with a rise in employees as reflected in Figure 2. In 2006 the total number of employees within the industry was 812, and by 2020 skyrocketed to 13'773. The most notable growth occurred between 2017 and 2018, when the workforce expanded from 5'338 to 7'924, marking a 48% increase. This growth can be attributed to investments made within the industry; one example is *Activision Blizzard's* (United States-based) acquisition of Swedish mobile developer *King* (Dataspelsbranschen, 2018). Further, Dataspelsbranschen (2018) shares that larger projects increased the demand for labor, such as *EA DICE*'s *Battlefield V* published in 2018. Given the overall growth in the total number of employees, it is reasonable to anticipate a corresponding increase in the number of female employees. Figure 2 illustrates the number of female employees in the Swedish video game industry, showing a remarkable growth from 107 in 2006 to 3,031 in 2020—an increase of nearly 3,000%. As previously mentioned, the total number of employees increased significantly over

time, however, Figure 2 visually illustrates a divergence between total employees and female employees suggesting a difference in growth rates.

Figure 2. Total and Female Employees in the Swedish Video Game Industry 2006-2020



Source: Dataspelsbranschen

The results are plotted in Figure 3 to provide a better visualization of female representation within the industry highlighting the unequal proportion of women in the workforce. In 2006, women made up 13.2% of the total employees in the Swedish video game industry, suggesting a gender-imbalanced workforce. This distribution declined to 9.9% in 2009, the lowest share experienced in the industry. The small decrease from 2019 and 2020 can be interpreted with Figure 2 where the total number of employees increased, suggesting that most of these new employees were male. Despite this, it can be said that the female distribution exhibited a gradual annual increase, reaching 22% by 2020. These changes over time suggest an increase in the proportion of women working in the video game industry and a step closer to equal representation. Figure 3 reflects that despite the increasing number of female employees suggested by Figure 2, women represent a small proportion of the total workforce.

Figure 3. Female Distribution of Total Employees
2006-2020



Source: Dataspelsbranschen and author's calculations

In Figure 3 there is a steeper increase in the female distribution between 2010 and 2011, from 10% to 14%. This increase can be attributed to the success of mobile games which engaged a more diverse and larger audience. The success of mobile games increased labor demand for both technical and managerial positions (Dataspelsbranschen, 2011). This is reflected in the workforce composition, particularly within mobile development companies as in 2011, mobile developers employed the highest proportion of women (Dataspelsbranschen, 2011). *King* continued this trend, reporting that women comprised 40% of their total hires in 2019 (Activision Blizzard, 2020). Overall, the dynamic growth of the Swedish game industry expanded its audience reach and generated new employment opportunities. However, as noted by Dataspelsbranschen (2011), the industry expressed a demand for technical and managerial roles, prompting further exploration into the skills provision and positions occupied by women.

4.2. The Role of Higher Education

According to Dataspelsbranschen (2017), one of the causes of low diversity within the industry is the skill shortage which is directly provided by education. Given that higher-level education does

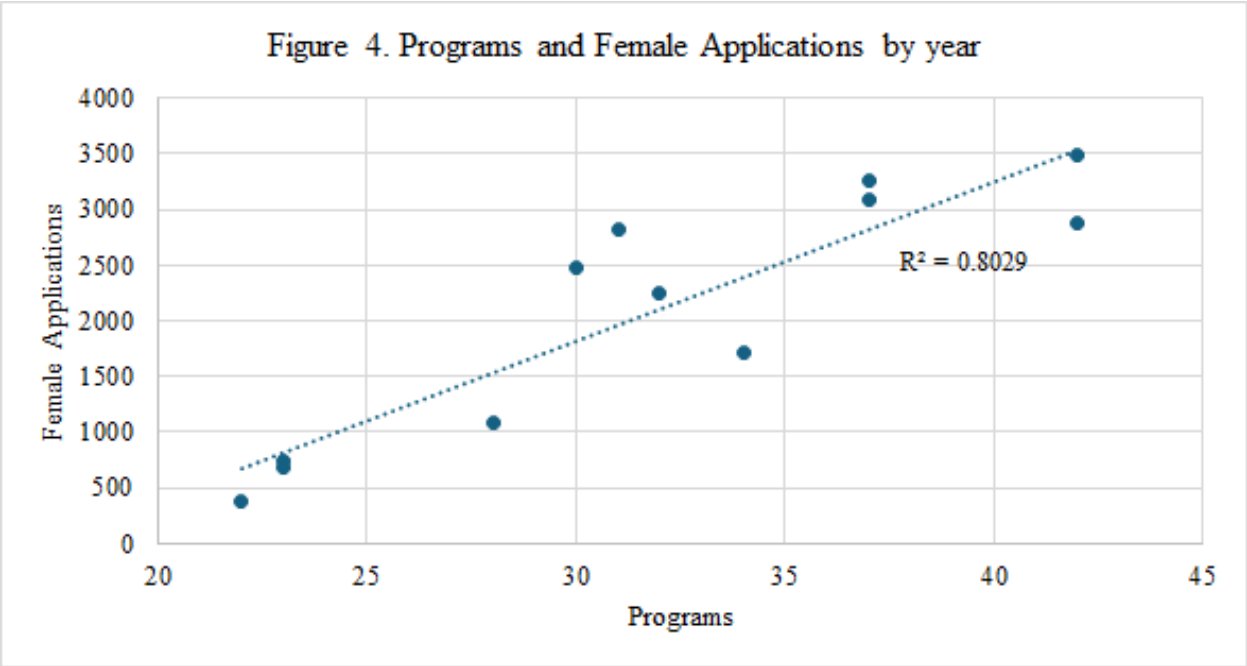
not have a cost in Sweden, one would expect an equal gender distribution in high-level education applications, as the high cost of studying can be a barrier when entering the video game sector (Nordic Council, 2017). Using the code displayed in Appendix B, it was possible to create an average on the share of female applicants per year for all selected higher education programs aimed towards the game industry, for which the results are shown in Table 1 below. The average distribution of female applications increased steadily and significantly, as out of all the applications to programs related to the game industry in 2008, 8.94% were female, whilst in 2019, the share was 27.72%. Despite this increase, out of all applicants, women continued to be the minority in game-related programs in Sweden. Table 1 further shows the average female distribution of those students accepted to game-related programs. In 2008, out of all students accepted, 13.69% were women, whilst by 2019, 30% were women, with the biggest increase between 2010 and 2011. This shows a constant increase in the share of women being accepted into their desired career. According to the *Swedish Game Education:2001-2016* (2017) report, female representation is greater for first-hand applications than it is for enrollment. Although students being accepted to high education programs does not imply, they will enroll in the program, in this case, there was a higher female representation in students accepted than in those applying.

In Europe, there has been a call to make STEM careers more attractive to women to increase gender diversity (Directorate-General for Research and Innovation, 2018) resulting in different countries investing in increasing the number of available programs within game education to attract more applicants. For instance, according to the *Game Education Index* (2017), in 2017 the Nordic countries offered around 65 higher education programs aimed at the game industry. These programs vary in emphasis including level design, game design, game writing, game animation, game programming, engine programming, 2D game art, 3D game art, animation, tech art, SFX, production management, quality assurance, and business analytics. As shown by Table 1, the number of higher education programs in Sweden aimed at the game industry have been increasing since 2008 from 22 programs to 42 programs by 2019. Sweden's investment in opening new programs may have caused the increase in the share of female applications, as universities will take in more students through more career paths, making it more attractive to those interested in game development.

Table 1. Average Share of Female Applications and Average Share of Women Accepted 2008 - 2019			
Year	Female Applications (%)	Women Accepted (%)	Total Programs
2008	8.94	13.69	22
2009	9.77	9.98	23
2010	9.92	10.24	23
2011	12.36	17.87	28
2012	14.31	18.97	34
2013	16.24	21.15	32
2014	17.03	21.49	30
2015	19.44	23.47	31
2016	21.55	25.91	37
2017	22.66	28.51	37
2018	24.58	27.43	42
2019	27.72	30.54	42
Source: *Universitets- och högskoler* and author's calculations			

Figure 4 below illustrates the number of programs and total female applications by year on a scatter diagram to further understand the impact of available programs on women's attempts to follow the gaming industry career path. From Figure 4 is clear that there is a close relationship between the number of programs and female applications, as years with fewer programs had fewer women applying. Further, Figure 4 also shows the trend line of such a relationship with an R-square value of 0.8029. The R-squared value suggests that 80% of the variance in female applications is

explained by the number of higher education programs available. This shows that increasing the number of higher education programs can help increase the number of women who aim to study a career directed at the game industry.



Source: *Universitets- och högskoler* and author's calculations

As per the *Nordic Game Education Index* (2017) and the *Swedish Game Education:2001-2016* (2017), programming has been one of the disciplines with the least amount of female representation as between 2011 and 2015, where 9.2% of the enrolled students were female. Table 2 below shows the total number of applications to any of the selected programs focused on programming (*Game Design and Programming*, *Programming in Game Development*, *Innovative Programming*, and *Game Programming*) and the number of female and male applications received. From Table 2 we can see that men apply to programming focused disciplines at a greater extent than women. In 2008, out of all 477 applicants, only 12 were women. This trend is maintained in consequent years, and by 2019 out of the 1'309 applications received, 172 were women. Although women represent a minority in the applications received, this shows that women's interest in programming careers has been increasing over time. Table 2 also shows that the number of programming disciplines available remained under five throughout time, suggesting

that women have shown an increasing interest in programming regardless of the number of programs available. Naturally, we cannot assume that all applicants were accepted into the program and finished the degree, for instance, between 2010 - 2015, on average, 75% of students enrolled in game education finished their careers and obtained a degree (Nordic Council, 2017). It is clear that women are more motivated to follow a career path within programming in game development as shown by the increasing number of applications, however, programming remains a male-dominated career.

Table 2. Female, Male, and Total Applications to Programming 2008-2019

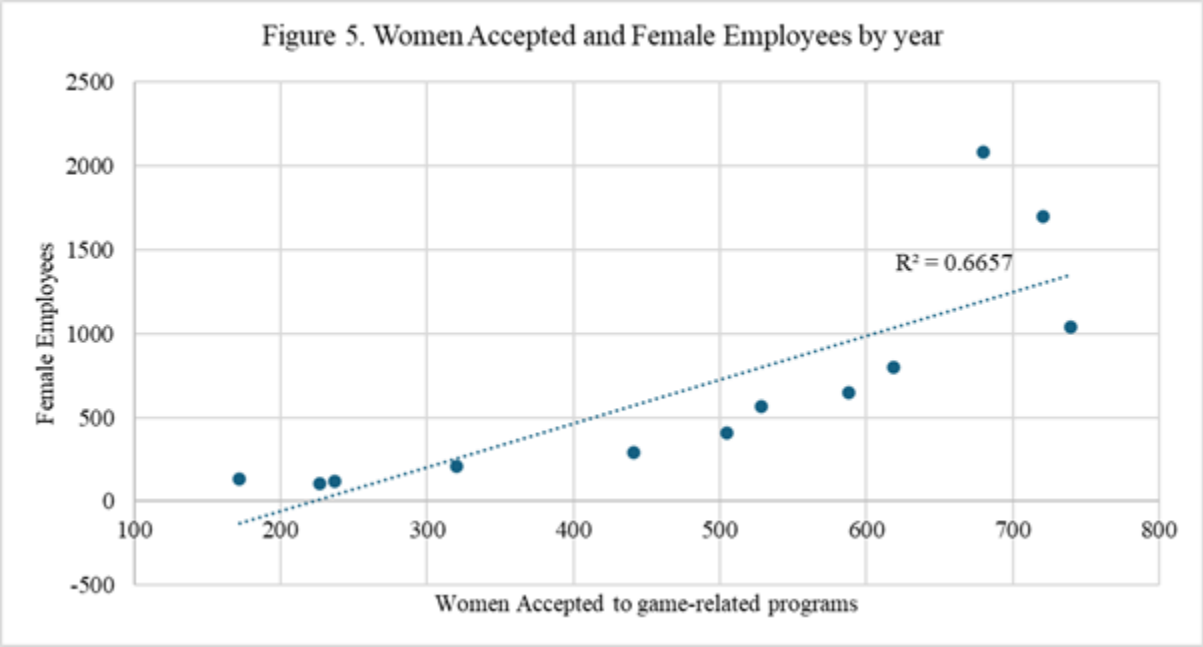
Year	Female Applications	Male Applications	Total Applications	Programs
2008	12	465	477	3
2009	42	863	905	4
2010	49	1000	1049	4
2011	74	1019	1093	4
2012	74	1090	1164	4
2013	89	1558	1647	4
2014	97	1681	1778	4
2015	114	1725	1839	4
2016	136	1748	1884	5
2017	220	1849	2069	5
2018	261	1728	1989	5

2019	172	1137	1309	4
Source: *Universitets- och högskoler* and author's calculations				

According to a 2016 survey by the International Game Developers Association (IGDA), 64% of all respondents had a college or university degree. A multiple linear regression analysis was carried out to test the hypothesis that a higher acceptance of women to higher education programs would lead to a higher number of women employed in the industry. The results are shown in Appendix C and can be interpreted in the following way; firstly, the F-statistics (2,9) have a value of 48.14, and a Prob>F value of 0.000, which indicates that the regression is highly statistically significant. Secondly, there is an R-squared value of 0.9145 from which we can infer that the number of women accepted, and year are good predictors of female employees, as 91% of the variation in female employees can be explained by the number of women accepted to programs and year. The model has also resulted in an adjusted R-squared value of 0.89, meaning that 89% of the variability in the dependent variable *female employees*, is explained by the dependent variables adjusted for the number of predictors in the model. The results also show that the coefficient of *accepted_female* has a value of -4.08744, this means that by a one-unit increase in women accepted, female employees is expected to decrease by 4.08, assuming all factors remain constant. The value P>|t| = 0.014 indicates a statistical significance at the typical 5% significance level, similarly, the results show a 95% confidence interval, confirming the statistical significance of the model.

For a visual representation, Figure 5 is a scatterplot with a fitted regression line for *female employees* and *female applications* - not including *year*. Overall, Figure 5 illustrates a statistically significant relationship between women being accepted to game-related programs and the number of female employees in the game industry's workforce. The R-square value for this simple linear regression is 0.6657, suggesting that about 66% of the variation in the number of female employees can be determined by women accepted. This is significant in the actual context of the video game industry as it could suggest that women applying and being accepted in a game-related education

(and eventually graduating) may result in an increase in female distribution in the video game industry.



Source: *Universitets- och högskoler* and author's calculations

Dataspelsbranschen (2024) shares that educational initiatives take time to have an impact on the labor market as most educational programs have a two or three-year duration. Although Figure 5 suggests a relationship between education and employment in the video game industry, it is crucial to understand that education is a long-term investment, as graduates are expected to enter the game industry a few years after their initial enrollment in game-related programs. Despite this, Dataspelsbranschen (2023) claims that the number of degrees from universities and colleges exceeded the number of job vacancies up until 2012. For instance, in 2015, 575 people graduated with a game education degree whilst the average of new hires between 2015 and 2021 was 750. It is unclear if women and men are hired at the same rate, however, the Directorate-General for Research and Innovation (2018) highlights that at the EU level, women continue to have a lower share of total employment than men even if they have attained the same level of education.

It is also common for employees to have a "non-traditional" education, such as self-taught learning or online courses (Ecorys & KEA, 2023). Dataspelsbranschen (2020) has highlighted the need for

supplementary educations that help potential participants acquire the needed skills to enter the game industry. In Sweden, there have been educational initiatives such as DONNA and the Valkyrie Game Jam aimed at increasing female distribution in the video game industry by providing key skills, competencies, and networking to women and non-binary people interested in technical careers (Sweden Game Arena, 2024). At European level, the initiative EU Code Week launched in 2013 aiming to promote interest in programming. In 2019, the event had 4.2 million participants of which 49% were women (Directorate-General for Research and Innovation, 2021). Initiatives such as the ones mentioned inspire women to join the game industry and could help with the training of aspiring game developers, although the extent of their impact on women's representation in Sweden has to be further researched.

4.3. Occupational Segregation

It is important to understand the extent of occupational segregation in Sweden. Table 3 shows the share of women in various occupations in Sweden in the years 2014, 2017, and 2020. Women represent most of the workforce for occupations such as *Nurses* and *Office Assistants & Secretaries*, with 87% and 82% respectively. Women's overrepresentation in these two occupations remained at the same level over the years. *Personnel and HR Managers* had a slightly high female representation of 62% in 2014, which increased to 66% by 2020. In contrast, women remained underrepresented as in *IT System Developers* consistently making up 22% of the workforce across all years. The occupation of *Engineers and Technicians* increased slightly throughout time, as women made up 17% of the workforce in 2014, 19% in 2017, and 22% in 2020. This distribution is close to that of the *Gaming Industry*, also shown in Table 3. The *Gaming Industry* had a female share of 18% in 2014, which increased to 22% by 2020. Despite this increase, it is clear that women have been underrepresented in STEM occupations, and overrepresented in healthcare and administrative occupations, suggesting the presence of occupational segregation in Sweden.

Table 3. Share of Women in Various Occupations (%)			
Occupation	2014	2017	2020

Gaming Industry	18%	19%	22%
Engineers and Technicians	17%	19%	20%
IT and System Developers	22%	22%	22%
Personnel and HR Managers	62%	65%	66%
Office Assistants and Secretaries	82%	82%	88%
Nurses	87%	87%	88%
Source: SCB and author's calculations			

According to Ecorys & KEA (2023) measuring and acknowledging occupational segregation within the video game industry is challenging due to the industry's "multi-disciplinarity" characteristic. My Nilsson is a researcher who in 2021 carried out independent research via LinkedIn to get an overview of occupational distribution in Swedish game studios, My Nilsson's findings are illustrated in Table 4. In 2021, the smallest share of women was found in the *Art* and *Design* departments with 8.5% and 16.7% respectively. The highest female share was in *Management* positions, in which 35% were women. One explanation offered by Dataspelsbranschen (2020) is that many studios highlight the importance of recruiting personnel beyond game development including marketing, accounting, support, testing, and human resources. My Nilsson's findings on Table 4 can be compared to those of Ecorys & KEA (2023) and Ashcraft et al (2016), who show similar results. Ashcraft et al (2016) report that in 2015, women made up 25% of all computing occupations, whilst Ecorys & KEA (2023) share that only 30% of all European game developers were women. We can then infer that from the results of Table and Table 4, the Swedish video game industry likely experiences occupational segregation.

Table 4, (adapted from Nilsson, 2021). Share of Women by Occupations in Swedish Game Industry	
Non-development	22.3%
Sound	18%
Art	8.5%
Tech	20%
Design	16.7%
Management	35%
Source: Nilsson, M. (2021)	

5. Discussion

The aim of this dissertation was to explore women in the Swedish video game industry. The outcome of this research aligns with the findings of previous research on international developer studios, as women remain underrepresented in the Swedish video game industry. The Swedish game industry has experienced a growth in the number of companies increasing the demand for labor. While this growth is encouraging, it is evident that women remain underrepresented in the video game industry as they still account for less than one-third of the industry's workforce. Similarly, the results show that the 2019 increase of employees lowered women's proportion, suggesting more male candidates were hired. Further research on the hiring rates of women applying to the video game industry is essential to ensure that recruitment processes are conducted fairly and without bias.

As shown in the analysis, there has been an increase in the number of women employed by Swedish game studios, especially after the success of mobile games. The success of mobile games in 2010-2011 not only meant for a higher workforce, but it also reached new audiences. According to the theory proposed by Weststar and Marie-Josée (2018), women's increased exposure to games may have influenced their professional trajectories, as they are likely to pursue careers aligned with their hobbies. Women that found themselves enjoying mobile games may consider a career path where they can contribute to the development of similar games. The higher increase in applications to game-related programs in 2010 - 2011 could have been the result of this, however, there are other factors to consider that were not included in the analysis.

The findings of this research suggest that women's underrepresentation in the video game industry can be the result of their small share in game-related education. The *Results & Analysis* section has shown that women actively seek to be part of game-related programs, although they make up for a small population of applicants and accepted students compared to men. The data shown in this research is consistent with the major trends by previous researchers such as Clark et al (2021) and Blickenstaff's (2005), who suggest that women may feel less confident about following a game-related career knowing that they will likely be a minority in the program and that they will remain a minority in any future employment in the video game industry compared to men. This can be interpreted as a cycle where not many women seek game-related careers due to their underrepresentation in both education and the workforce, making game-related programs and jobs less attractive for women, further decreasing female representation in the industry, and so on. This is significant as it highlights the need of initiatives that can address both education and the visibility of the few women in the industry.

As suggested by the Game Education Index (2017) there is very little gender distribution in programs focused on programming, sound, and design. The results presented in this thesis align with those of the Game Education Index (2017) and those of Blickenstaff (2005) with women being a minority in the programming-focused disciplines. Despite the low representation of women in programming careers, this study has identified an increasing interest in video game careers shown by an increasing number of women applying and being accepted to programming disciplines. This is significant given that, as shown in the previous section, the number of

programming disciplines offered in Sweden had minimal change. Therefore, initiatives like the EU Code Week could provide further inspiration and training to women that are interested in programming or choose to not enroll in formal education, especially since, the number of applications to high education programs usually exceeds the number of places available (the number of applicants for the country's game education programs is large and exceeds the number of education places many times over (Dataspelsbranschen, 2023).

Another approach to women's underrepresentation, can be increasing the number of higher education programs offered in Sweden. As shown in the *Results &* Analysis section, the increase in the number of game-related programs in Sweden was accompanied by an increase in women's applications. *The Swedish Game Education: 2001*-2016 (2017) share that, apart from the Nordic Council initiative, the increase in programs could be the division of one program into several programs. Regardless, the results of this research is significant since as Dataspelsbranschen (2023) has suggested, the industry has been experiencing a shortage of skilled labor. However, as pointed by the *Swedish Game Educaiton:2001-2016* (2017) report, it is key to ensure the quality of such programs making sure that they include the competencies needed for game development. An increase in suitable and relevant game-related programs may be helpful in increasing women's applications.

The regression and scatter plot presented in the analysis provided a good insight into the impact of women's acceptance to higher education programs on employment numbers. It is important to highlight that the results do not account for the length of educational programs. This means that for the analysis to be applicable, one should account for the length of various educational programs, as they usually take two to three years to complete. Similarly, although the results of this research show a moderate variance in employees by the number of women accepted, the analysis does not enable us to prove that women taking a game-related program will be hired into the industry. Dataspelsbranschen (2024) has highlighted that a key player in entering the game industry is the provision of structured internships within educational programs. Thus, then calls for an examination of the syllabus and relevance of the offered programs and, if needed, modifying them to incorporate an internship. *She Figures* (2018) report that women have a lower employment share than men. As internships within educational programs are one of the most direct ways to

enter the video game industry (Dataspelsbranchen, 2024), women may feel encouraged to apply to these programs as they increase their employment opportunities.

This research has shown that an equal country such as Sweden, can still face occupational gender segregation. The findings presented align with past research carried by Mandel and Semyonov's (2006) and Lemons and Parzinger (2001), who point that women are overrepresented in healthcare and administrative occupations, often in the public sector, whilst underrepresented in STEM occupations. There has been an improvement over time, with the share of women gradually increasing, however, as shown in the *Results and Analysis*, these improvements are minimal.

One of the main findings of this study has been the lack of information within the Swedish video game industry's occupational distribution and its implications. Only one other study, to our knowledge, has examined occupational segregation in the video game industry – My Nilsson's 2021 research. Although My Nilsson's research is outside the time scope of this thesis, it reinforces the argument that game-development roles such as art, design, audio, and programming are heavily male-dominated as shared by the IGDA (2016) and by Bailey et al (2019). The lack of information on the occupational distribution in the Swedish video game industry poses a significant challenge, especially given that occupational segregation can influence young people's career choices – as shared by Miller et al (2004). Women may assume there is occupational segregation in the Swedish video game industry similar to international studios, and similar to the national occupational distribution, making women less confident in following a career in game development. Furthermore, women may also choose to not join the video game industry as they may assume their only career path within the industry is in managerial positions with little impact and influence on game development. This research then raises the question of whether sharing the extent of occupational segregation in Swedish developer studios can have a positive impact on women's pursuit of developmental careers and, ultimately, their equal participation in the Swedish video game workforce.

Dataspelsbranschen (2023) and the Directorate-General for Research and Innovation (2018) have highlighted challenges in collecting occupational distribution data. These challenges include accounting for the status and reports of independent and solo developers, who, as demonstrated in this paper, make up the majority of developer studios in Sweden. Regardless, this is not a challenge

for major developers in the Swedish game industry with a bigger employee pool, who can still attempt to gather data and share it with the public. This presents an opportunity for game studios on all scales to be transparent in their occupational distribution by gender, as this would create a complete overview of a social issue that could be further explored and, hopefully, addressed.

Several questions remain to be resolved, in particular what approach should be taken to increase the share of women in the video game industry. This research has identified potential areas of improvement that could enhance the understanding and approach to addressing gender disparities in the Swedish video game workforce. Ultimately, the outcomes of this research have provided insight into women's representation in the video game industry. However, the results should be interpreted with the research scope and limitations in mind.

6. Conclusion

This research aimed to explore women's representation in the Swedish video game industry through both qualitative and a quantitative analysis, examining aspects from education to occupational distribution. The study was guided by the following research questions:

Research Question 1.

How has the share of women changed over time in the Swedish game industry?

Research Question 2.

What is the relationship between education and women's underrepresentation in the video game industry?

Research Question 3.

To what extent is occupational segregation present in the video game industry and what are its implications?

In response to the first question, this study has found that women have been underrepresented in the video game industry. The data shared shows that women make up less than a third of the workforce in the Swedish video game industry. This finding reflects a persistent gender imbalance,

aligning with broader trends observed internationally. Such underrepresentation limits the diversity of perspectives in game development and perpetuates structural barriers for women in the industry and for those wanting to join the industry.

Regarding the second research question, the quantitative analysis highlighted the significant role of education in the road to equal representation in the Swedish video game industry. The availability of game-related programs emerged as reliable predictors of women's interest in and applications to such programs. The data suggests that increasing women's participation in these educational disciplines could lead to a higher share of women in the video game industry. However, the small sample size of this research limits the generalizability of the findings. Despite this, this thesis offered valuable insights into how restructuring and enhancing game-related disciplines can influence women's representation in the Swedish video game workforce.

The third research question remains less resolved, considerably more work will need to be done to fully address the extent of occupational segregation in the Swedish video game industry. While this study did not conclusively confirm occupational segregation within Swedish game developer companies, it did provide evidence of its existence at a national level. The results further point to the possibility of occupational segregation following trends observed in international contexts. The major finding when addressing this question was the lack of information on the occupational distribution in Swedish studios. This raises important questions about the implications of this gap and the impact that occupational segregation may have on women's professional careers and the overall inclusivity of the industry. Addressing these questions will require further research with larger sample sizes, more detailed data, and significant support from game studios.

While this study focused on individuals who identify as male or female due to limitations in scope, future research should aim to address the experiences of non-binary individuals to provide a more comprehensive understanding of gender representation in the Swedish video game industry. Additionally, exploring the extent of sexist culture in Swedish game studios would offer valuable insights into the working conditions and their impact on women's representation. Research into the gender wage gap and its effects on the Swedish video game industry would also complement the findings of this study, further enriching the conversation around gender equality in the industry.

In conclusion, this study contributes to understanding the barriers to gender equity in the Swedish video game industry while identifying actionable pathways to improvement. Promoting game-related education for women and investigating occupational segregation further are essential steps toward a more inclusive industry. These findings not only enhance awareness but also provide a foundation for future research and policymaking aimed at fostering gender balance in the workforce.

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Appendix A. Summary of Datasets

A1. Summary of Dataset 1

Variable	Obs	Mean	Std. dev.	Min	Max
year	381	2014.186	3.393403	2008	2019
program_id	0				
school	0				
applied_male	381	320.7323	222.8235	3	1208
applied_fe~e	381	65.1811	56.26382	0	275
applied_to~l	381	385.9134	255.1991	4	1373

A2. Summary of Dataset 2

Variable	Obs	Mean	Std. dev.	Min	Max
year	378	2014.148	3.391245	2008	2019
program_id	0				
school	0				
accepted_m~e	378	69.33862	53.16943	0	269
accepted_f~e	378	15.28571	15.18403	0	91

Appendix B. Code of Do-File

```
cd "C:\Users\angel\OneDrive\Desktop\School\Thesis Stata"

import excel "C:\Users\angel\OneDrive\Desktop\School\Thesis Stata\for_Stata.xlsx", sheet("A1") firstrow clear
save "A1.dta", replace

import excel "C:\Users\angel\OneDrive\Desktop\School\Thesis Stata\for_Stata.xlsx", sheet("A2") firstrow clear
save "A2.dta", replace

/////////////// Programs and Applications

use "A1.dta", clear

gen percent = (applied_female/applied_total)*100
tabstat percent, by(year) statistics(mean sum count)

tabstat applied_female if program_id == "Game Design and Programming" | program_id == "Programming in Game
Development" | program_id == "Innovative Programming"| program_id == "Game Programming", by(year) statistics(
mean sum min max count)

tabstat applied_male if program_id == "Game Design and Programming" | program_id == "Programming in Game
Development" | program_id == "Innovative Programming"| program_id == "Game Programming", by(year) statistics(
mean sum min max count)

tabstat applied_total if program_id == "Game Design and Programming" | program_id == "Programming in Game
Development" | program_id == "Innovative Programming"| program_id == "Game Programming", by(year) statistics(
mean sum min max count)

egen avg_percent = mean(percent), by(year)
bysort year: keep if _n == 1

gen program_indicator = 1
collapse (sum) applied_female (count) num_programs=program_indicator, by(year)

/////////////Accepted and Employees


    use "A2.dta", clear

     bysort year: gen num_programs = _N
     collapse (sum) accepted_female , by(year)
     gen female_employees = .
replace female_employees = 135 if year == 2008
replace female_employees = 109 if year == 2009
replace female_employees = 121 if year == 2010
replace female_employees = 212 if year == 2011
replace female_employees = 293 if year == 2012
replace female_employees = 405 if year == 2013
replace female_employees = 563 if year == 2014
replace female_employees = 651 if year == 2015
replace female_employees = 800 if year == 2016
replace female_employees = 1041 if year == 2017
replace female_employees = 1699 if year == 2018
replace female_employees = 2079 if year == 2019

    list

    regress female_employees accepted_female year
```

Appendix C. Results of Multiple Regression Analysis

Source	SS	df	MS	Number of obs	=	12
				F(2, 9)	=	48.14
Model	4146887.02	2	2073443.51	Prob > F	=	0.0000
Residual	387665.644	9	43073.9605	R-squared	=	0.9145
				Adj R-squared	=	0.8955
Total	4534552.67	11	412232.061	Root MSE	=	207.54

female_employ~s	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
accepted_female	-4.08744	1.344327	-3.04	0.014	-7.128519	-1.046361
year	383.6313	74.95957	5.12	0.001	214.061	553.2016
_cons	-769797.8	150301.5	-5.12	0.001	-1109803	-429792.3